Machine Learning and Artificial Intelligence Applied to Computational Arts, Music, and Games

Kıvanç Tatar

CHALMERS UNIVERSITY OF TECHNOLOGY | UNIVERSITY OF GOTHENBURG

These slides are live at:

https://ktatar.github.io/2026-03-docent-lecture/

Kıvanç Tatar

CHALMERS UNIVERSITY OF TECHNOLOGY | UNIVERSITY OF GOTHENBURG

Research Themes

-Deep Learning and Audio Technologies

-AI andin Sound and Music Interactions

-AI in Computational Creativity and Game Design

-Societal Impact of AI in Culture, Arts, and Music

Deep Learning and Audio Technologies

Focuses on technical innovations in sound synthesis and modeling using deep learning:

- Variational Autoencoders (VAEs)
- Neural Fourier operators for timbre and acoustics.
- Real-time audio generation from raw data.
- Applications in musical acoustics and sound design.

Deep Learning and Audio Technologies

- Latent Timbre Synthesis
- RawAudio Variational Autoencoder
- Raw Music from Free Movements
- Reinforcement Learning for Musical Performances with Moving Machine

Latent Timbre Synthesis

```
<iframe width="560" height="315"
src="https://www.youtube.com/embed/ZJm-N_-ySe0?
si=WDxjA7frj4Jj3COq" title="YouTube video player" frameborder="0"
allow="accelerometer; autoplay; clipboard-write; encrypted-media;
gyroscope; picture-in-picture; web-share" referrerpolicy="strict-
origin-when-cross-origin" allowfullscreen></iframe>
```

Latent Timbre Synthesis

Kıvanç Tatar

This is a table

Activities				
Lectures	Introduction to Art and Technology	Introduction to AI and ML	Creativity, Group work, and Tools for Innovation	
Tutorials	Creative Coding with Sound (PureData)	Multimedia Design with TouchDesigner	Deep Learning for Multimedia	Interactive ML with Physical Computing
Project	Project Proposal	Design Iterations	Final Prototype	Reflections

Text border size is set in section class *padding: 5%* in my_theme.css

This is the same table without borders

Activities				
Lectures	Introduction to Art and Technology	Introduction to AI and ML	Creativity, Group work, and Tools for Innovation	
Tutorials	Creative Coding with Sound (PureData)	Multimedia Design with TouchDesigner	Deep Learning for Multimedia	Interactive ML with Physical Computing
Project	Project Proposal	Design Iterations	Final Prototype	Reflections

See how *Activities* are not bold now. That is because th class is updated in section.no_border class in my_theme.css

Two column slide

Originally solution from https://github.com/orgs/marp-team/discussions/192#discussioncomment-1516155

Can't believe this works. YOLO

If you are brave for more sections, here is an online tool for css grids: https://grid.layoutit.com/

Image on the left

Kıvanç Tatar

CHALMERS UNIVERSITY OF TECHNOLOGY | UNIVERSITY OF GOTHENBURG

AND Image on the right

Kıvanç Tatar

CHALMERS UNIVERSITY OF TECHNOLOGY | UNIVERSITY OF GOTHENBURG

Image aligned on the left

Image aligned on the right

Image aligned at the center

Local videos are possible

Online videos as well

```
<iframe width="560" height="315"
src="https://www.youtube.com/embed/nx2Nj3I7NyU?si=5D-
PemuLUUt3qepd&controls=0" title="YouTube video player"
frameborder="0" allow="accelerometer; autoplay; clipboard-write;
encrypted-media; gyroscope; picture-in-picture; web-share"
referrerpolicy="strict-origin-when-cross-origin" allowfullscreen>
</iframe>
```

This should work only when the presentation is hosted online. Youtube does not allow local hosts apparently.

Fancy image options

`<br/>` command works for creating empty spaces [2].

[1] this is a smaller text. Great for citations.

[2] https://marpit.marp.app/image-syntax

Inline HTML commands works

Text can be changed for example

BIG TEXT

How to export to github pages?

- Copy this template
- Setup github pages to deploy from a branch. Choose main branch.
  - There is no workflow, so that what you see local is what you get online as a static html.
- Once you are done with your presentation, render it to html locally.
  - On VS Code, you can just use the marp icon on the top right.
  - Make sure the file saved is index.html.
- Commit changes to the remote repo.
- Done! Page should be online at [user-name].github.io/[repo-name] after a few minutes.

Latex Math friendly

Inline math

Render inline math such as $ax^2 + bc + c$ [1].

Math block

$$I_{xx} = \int \int_R y^2 f(x, y) \cdot dydx$$

$$f(x) = \int_{-\infty}^{\infty} \hat{f}(\xi)\, e^{2\pi i \xi x}\, d\xi$$

[1] More details: https://github.com/marp-team/marp/blob/main/website/docs/guide/math-typesetting.md

Videos need to be formatted for web

You can use ffmpeg locally for this.

```
ffmpeg -i input.mp4 -c:v libvpx-vp9 -crf 30 -b:v 0 output.webm
```

note: tested and works

[1] Source: https://jshakespeare.com/encoding-browser-friendly-video-files-with-ffmpeg/

FFMPEG continued: another solution

fast and efficient h264 for the web :

```
ffmpeg -i in.mp4 -c:v libx264 -profile:v high -preset slow -crf 24 -c:a aac -b:a 96k -movflags +faststart out.mp4
```

More efficient but slower with VP9/webm

```
ffmpeg -i in.mp4 -c:v libvpx-vp9 -crf 33 -b:v 0 -c:a libopus -vbr on -b:a 64k out.webm
```

[1] https://www.reddit.com/r/ffmpeg/comments/8kxjhz/encoding_video_for_my_website/

FFMPEG continued

I haven't tried this solution yet.

WebM:

```
ffmpeg -i my-original-video.wmv -f webm -vcodec libvpx-vp9 -vb 1024k my-new-video.webm
```

MP4:

```
ffmpeg -i my-original-video.wmv -vcodec libx264 -f mp4 -vb 1024k -preset slow my-new-video.mp4
```

[1] Source: https://jshakespeare.com/encoding-browser-friendly-video-files-with-ffmpeg/